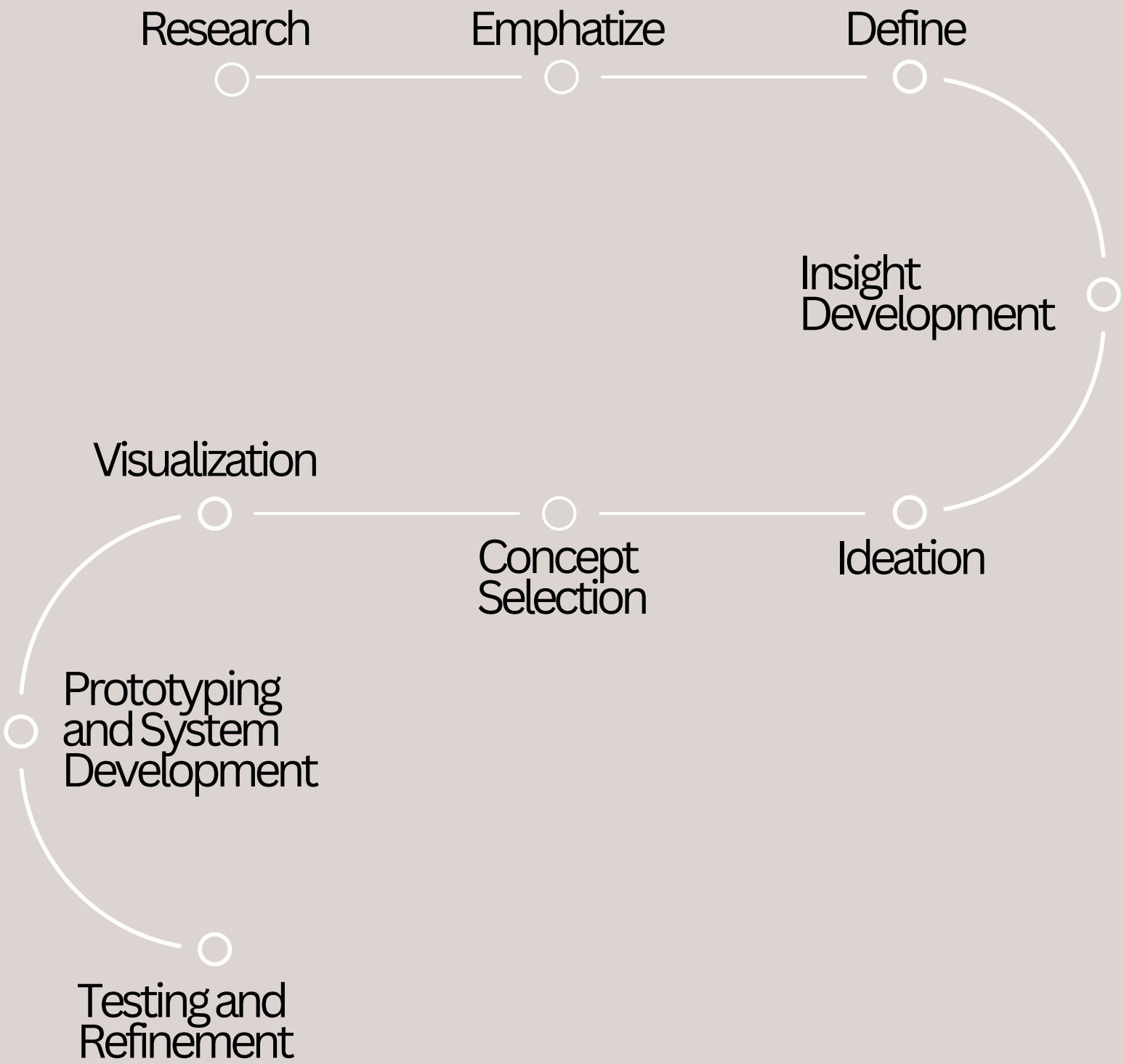
Design Milestones

Multi-Cezve Turkish Coffee Preparation Device



Damla Genel S039857

Research

Keywords: User research, coffee ritual, hosting culture, observation, interview, social interaction

Selected Impactful Prompt: “Can you help me prepare short interview questions for a user who sees making coffee as less about the drink itself and more about creating a warm shared moment with friends?”

AI Role: Clarifier

AI Response Summary: ChatGPT generated a structured set of interview questions focusing on social rituals, emotions, and user interaction around coffee preparation. Instead of focusing only on the machine itself, the questions explored topics such as hospitality culture, the role of coffee in social gatherings, and the emotional meaning of preparing coffee for guests.

Reflection: This interaction helped frame the research phase more effectively. Initially, my focus was primarily on the technical aspects of coffee preparation. However, the AI-generated interview questions shifted my attention toward the social and cultural dimensions of the coffee ritual. This allowed me to investigate how people interact around coffee preparation and how hosting practices influence the experience.

Design Decision: The project focus was expanded to include social interaction and shared rituals, rather than purely improving the efficiency of coffee preparation.

Accepted:
Interview questions prepared with AI
Social-ritual focused research approach

Rejected:
Research focused only on technical coffee preparation

Transformed:
Interview responses and observations were transformed into user tensions and insights

Emphatize

Keywords: Persona development, user journey, emotional experience, hosting stress, behavioral insights

Selected Impactful Prompt: “What do you think will be best to select as the top tensions from my research findings?”

AI Role: Re-framer

AI Response Summary: AI helped analyze the research results and identify several tensions in the coffee preparation experience. The strongest tension identified was the conflict between the host’s desire to remain socially engaged with guests and the attention required to prepare coffee properly.

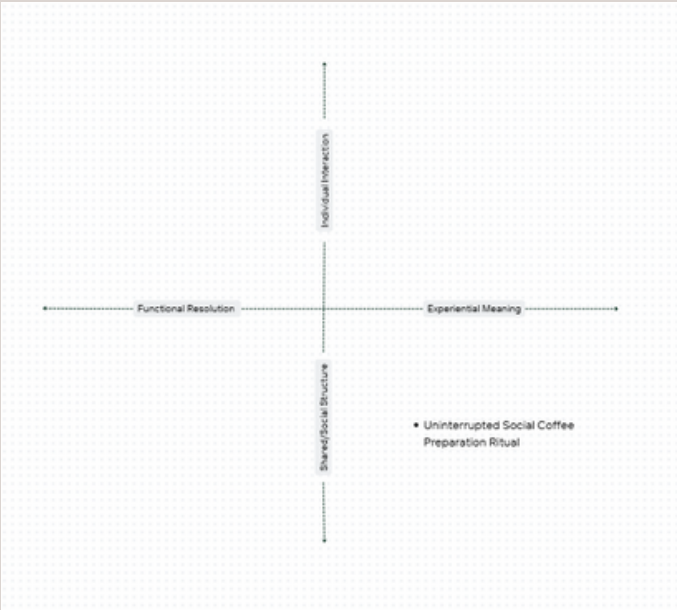
Reflection: AI’s suggestions helped reorganize the research findings into clear design tensions. Rather than seeing each observation as a separate issue, the process revealed an underlying conflict that shapes the entire experience.

Design Decision: The host wants to remain socially engaged with guests, but preparing coffee requires attention and interrupts the shared moment.

Accepted:
Primary tension definition

Rejected:
Minor tensions that did not strongly influence the concept

Transformed:
User observations were transformed into design insights



Define

Keywords: Problem framing, design insight, opportunity areas

Selected Impactful Prompt: “What would be the strongest tension to focus on for the project?”

AI Role: Clarifier

AI Response Summary: AI suggested prioritizing tensions that appear in multiple research sources, and that could realistically lead to a design solution. It helped refine the project problem statement by emphasizing the conflict between social interaction and preparation tasks.

Reflection: The clarification provided by AI helped ensure that the project addresses a meaningful user problem rather than focusing on superficial improvements.

Design Decision: The project was defined as designing a system that allows users to prepare coffee while remaining socially engaged with guests.

Accepted:
Social engagement as the central design goal

Rejected:
Solutions that focus only on increasing speed or automation

Transformed:
The tension was transformed into a clear project framing statement

Insight Development

Keywords: User insights, experience design, social rituals

Selected Impactful Prompt: “How can coffee preparation support social interaction instead of interrupting it?”

AI Role: Re-framer

AI Response Summary: AI suggested considering the waiting period and preparation process as part of the social ritual rather than as an interruption. It emphasized the importance of designing experiences that keep users engaged with guests during preparation.

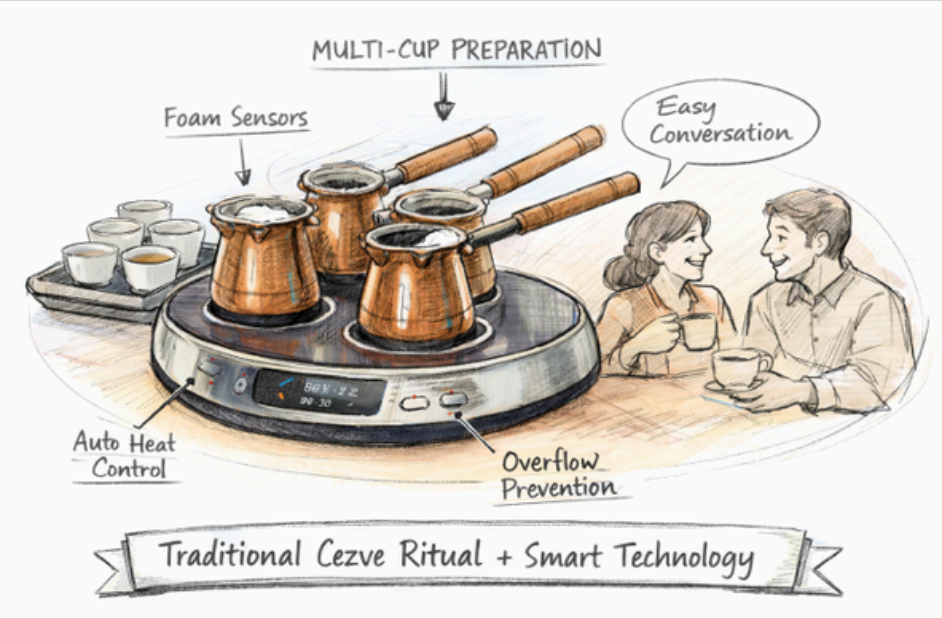
Reflection: This reframing shifted the design approach. Instead of removing the ritual through automation, the goal became supporting and enhancing the ritual while reducing stress.

Design Decision: The design should preserve the traditional Turkish coffee ritual while reducing the need for constant supervision.

Accepted:
Ritual-centered design approach

Rejected:
Fully automated coffee machine concept

Transformed:
Waiting time became part of the shared social experience



Ideation

Keywords: Concept exploration, sketching, brainstorming

Selected Impactful Prompt: “Could you transform these sketches into realistic product renderings?”

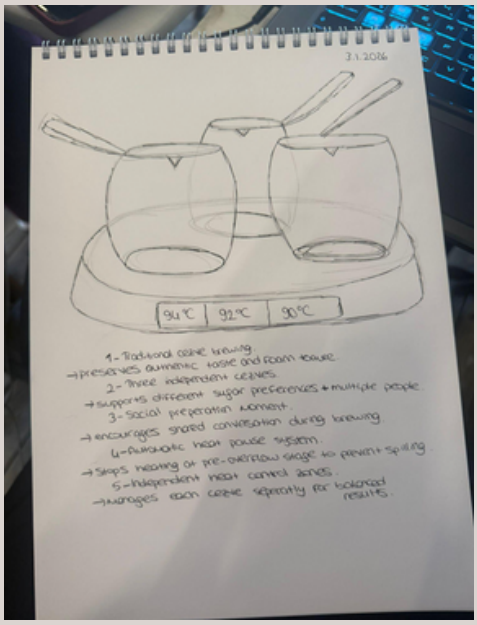
AI Role: Expander

AI Response Summary: AI generated visual interpretations of the sketches and explored different design directions for the coffee machine.

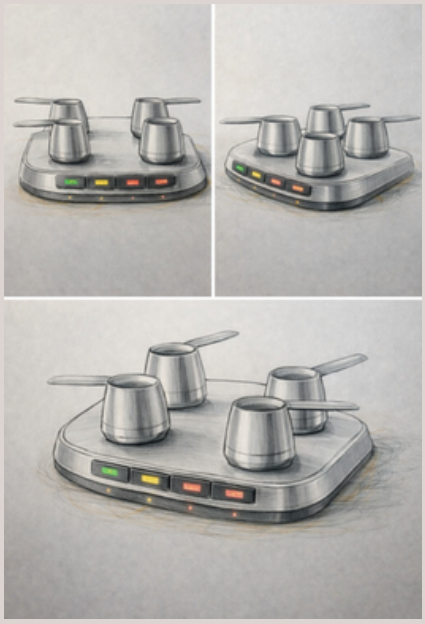
Reflection: The visualizations allowed me to evaluate the concepts more clearly. Seeing the ideas represented in a realistic way helped compare their potential impact and usability.

Design Decision: Develop a concept that enables multiple coffees to be prepared simultaneously.

Accepted:
Multi-cup preparation concept



Rejected:
Storage-focused coffee machine concepts



Transformed:
Rough sketches became refined visual concepts

Concept Selection

Keywords: Concept evaluation, strategic decision, design justification

Selected Impactful Prompt: “Why did I decide on my final concept?”

AI Role: Critique

AI Response Summary: AI analyzed the proposed concepts and evaluated them based on market conditions and the defined user persona. It suggested that the sensor-supported cezve system best addresses the core tension of divided attention.

Reflection: This critique helped justify the final concept with clear reasoning, ensuring that the decision was based on both user needs and product differentiation.

Design Decision: Select a concept that integrates sensor-supported overflow prevention while maintaining the traditional cezve preparation method.

Accepted:
Sensor-supported cezve system

Rejected:
Front-facing social machine concept

Transformed:
The concept evolved into a multi-cezve preparation device

Visualization

Keywords: Concept visualization, rendering, form exploration

Selected Impactful Prompt: “Make these sketches look more realistic.”

AI Role: Expander

AI Response Summary: AI generated realistic visualizations of the product, showing four cezves placed on a shared base with independent heating zones.

Reflection: These visuals helped communicate the concept clearly and evaluate the proportions, layout, and interaction of the product.

Design Decision: Use a shared base with four independent heating zones to allow simultaneous coffee preparation.

Accepted:
Four-cezve configuration

Rejected:
Single-cup preparation structure

Transformed:
Hand sketches into digital concept renders

Prototyping and System Development

Keywords: Interaction design, interface design, system logic

Selected Impactful Prompt: “How should the indicator system communicate the brewing status?”

AI Role: Clarifier

AI Response Summary: AI suggested using color-coded indicators and aligning the button layout with the position of the heating circles.

Reflection: These suggestions improved the clarity of the interaction design and made the system easier for users to understand.

Design Decision: Use color-coded indicators Green – coffee ready, Blue – brewing in progress, White – heating zone inactive

Accepted:
Color-coded feedback system

Rejected:
Text-only interface

Transformed:
Interface layout aligned with cezve positions

Testing and Refinement

Keywords: Design refinement, ergonomics, interface clarity, visual consistency, product coherence

Selected Impactful Prompt: “Fix the model so that all four cezves are the same size and proportions, then generate realistic renders of the product from multiple views while keeping the rest of the design unchanged.”

AI Role: Critique

AI Response Summary: The AI-generated renders revealed several visual and functional inconsistencies in the prototype. These included slight proportional differences between the cezves, unclear interface hierarchy, and placement areas that could be improved for usability. The visualizations helped identify areas where the design could be refined to create a more balanced and intuitive product.

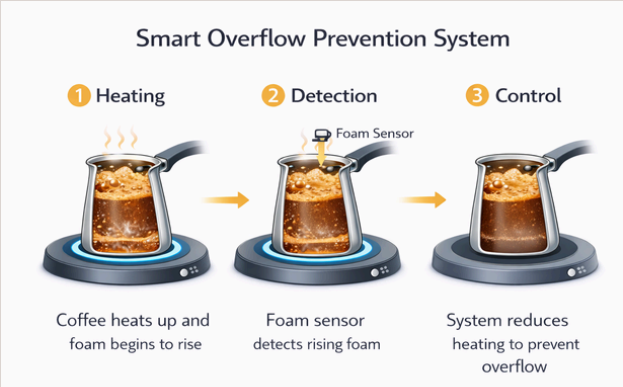
Reflection: During the refinement stage, I evaluated the visual balance, usability, and clarity of the product. The generated renders allowed me to observe the design as a whole system rather than as individual components. This helped identify several aspects that could be improved to strengthen both the aesthetic coherence and the user interaction of the device. Through this iterative process, the design evolved from a conceptual prototype into a more consistent and resolved product proposal.

Design Decisions:

Standardization of Cezves: All four cezves were redesigned to have identical proportions and size. This adjustment ensured visual consistency and created a more balanced and harmonious appearance for the overall device. Uniform components also help communicate that each heating zone functions in the same way.

Interface Reorganization: The control interface was reorganized to make the interaction clearer and more intuitive. A color-coded indicator system was defined to communicate the brewing status of each heating zone. This system allows users to quickly understand the status of each cezve at a glance without needing to read labels or instructions.

Lowered Heating Zones: The heating areas were redesigned to sit slightly below the surrounding surface. This subtle recessed geometry visually defines the heating area while still allowing the cezves to be placed freely. It helps guide placement without physically restricting the user’s movement, improving both usability and visual clarity.

Ergonomic Handle Improvement: The handle design of the cezves was refined to provide a more comfortable and secure grip. The updated geometry supports easier handling during pouring and reduces the risk of slipping, especially when the cezve is hot.

Unified Product Form Language: The base and the cezves were redesigned to follow the same modern design language. This created a stronger visual relationship between the components and made the product appear more cohesive as a single system rather than separate elements.